

TypeScript Foundations: Essentials for JavaScript Developers

Understanding the Why of TypeScript



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Our GitHub Repo

<https://github.com/FeynmanFan/pstypescriptcc>



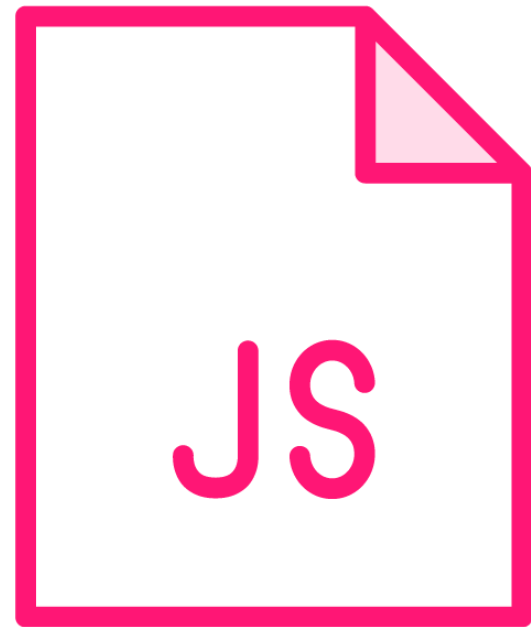
I Love TypeScript

I'm about 50/50 on Microsoft

I've been working with JavaScript
since the very beginning



How JavaScript Developed



Possibly the most successful programming language of all time

But it developed in an ad hoc way

Good, but also bad

TypeScript is an effort to place a design ethos atop JavaScript

Especially a type system – "Type"Script

Allow developers to leverage the language tools they know



So, Is TypeScript Just C# on top of JavaScript?

No

Another gem from
Anders Hejlsburg

TypeScript is
enhancing JS, not
supplanting it

"Pythonic"

TypeScript is
JavaScript-ic

Preserving what
design ethos there is



Does Typing Matter?

**"Static" or "Strong" Typing...not
really the same thing**

Focus on "type coercion"



Simple Type Coercion in JavaScript

```
var x = 5;  
x = "drum solo";  
console.log(x);
```



Type Coercion

Sometimes it's useful to coerce
to a new type

Strict typing is *commitment*

I think that type coercion is
often a code smell

Why else recommend const over var?

*Type coercion happens more
often due to an error in coding
than by design with loose typing*

[1, 2, 3]



Unit Testing and Typing

```
$employee.Birthday = "This is not a birthday and will break the code that tries to use it";
```

```
employee.Birthday = "This is not a birthday and will break the code that tries to use it";
```

```
employee.Birthday = "This is not a birthday and will break the code that tries to use it";
```

```
employee.Birthday = "2024-06-01";
```

```
employee.Birthday = DateTime.Parse("2024-06-01");
```

```
// Does TypeScript enforce strong typing?
```



TypeScript Type Enforcement

```
x = 5;  
x = "drum solo";
```

```
// TypeScript type enforcement happens at compilation time  
// only, and only for TypeScript code.
```



Compilation vs. Transpilation

The JavaScript emitted from processing TypeScript is at the same level of abstraction as TypeScript.





Version Controlling TypeScript and JavaScript



Keeping Binaries in Version Control Stinks

Craft your .gitignore

**Keep the junk out of
version control**

**Some stuff changes every time we
build – that's junk**

What is the principle?



**We version control only the
purely non-deterministic
elements of the system.**



Deterministic

$$346 \times 183 \\ = 63318$$

// Today, tomorrow, now and forever

// 63318 is the deterministic outcome of $346 * 183$



An Exception to the Rule

Partial migration

You can't just add *.js
to gitignore

The price of
the transition



Demo: How to Version Control TypeScript



Demo: Debugging TypeScript

